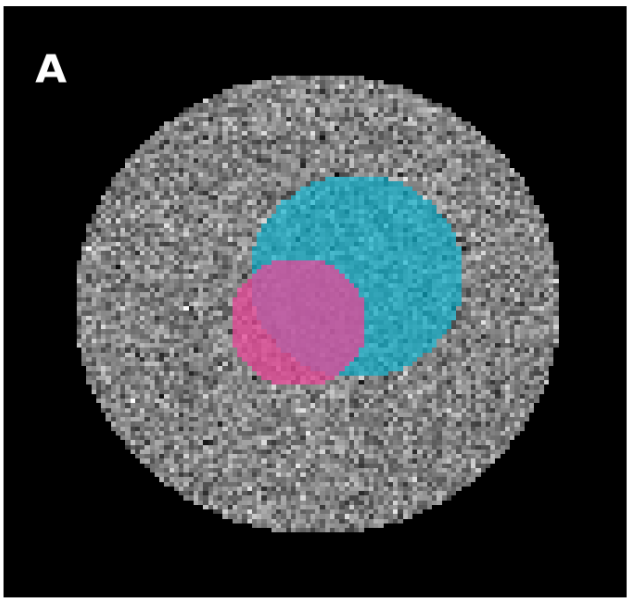
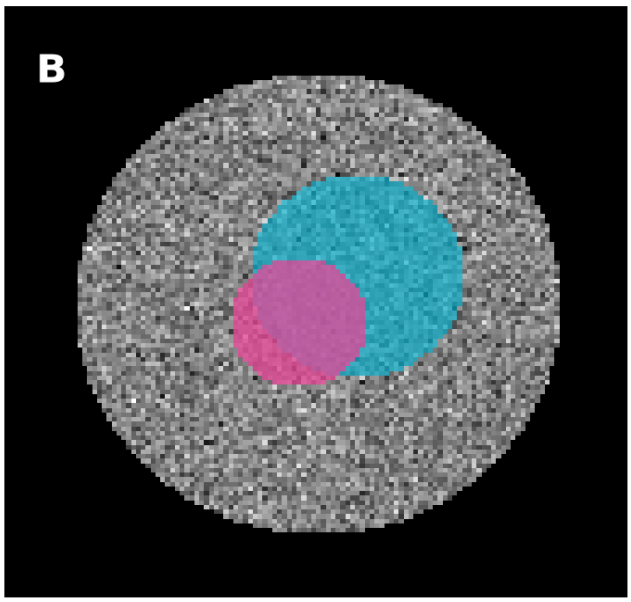
PhysioRANO NEURO-ONCOLOGY RADIOLOGY REPORT

Axial T1c (Post-Contrast)



Sagittal T2-FLAIR (Edema & Core)



CLINICAL HISTORY:

40-year-old Female presenting for post-radiotherapy Glioblastoma (GBM) evaluation at **45 days** post-treatment.
• Molecular Profile: IDH1/2 **Positive (Mutated)** | MGMT Promoter **Positive (Methylated)**.
• Current Status: Routine follow-up scan monitoring, mild headaches.

FINDINGS:

BRAIN PARENCHYMA & 3D TUMOR SUB-REGIONS:
Multi-modal 3D MRI quantitative volumetric analysis demonstrates:
• **Whole Tumor (WT) Volume: 24.50 cm³** (includes surrounding T2-FLAIR signal hyperintensity).
• **Tumor Core (TC) Volume: 12.20 cm³**.
• **Contrast-Enhancing Tumor (ET) Volume: 18.50 cm³**.

BIOPHYSICAL TUMOR GROWTH TRAJECTORY:
Biophysical neural network modeling evaluates the spatial invasion and proliferation dynamics:
• **Tissue Diffusion Speed:** Low spatial diffusion coefficient (**0.1245 mm²/day**).
• **Cell Proliferation Rate:** Modest cell proliferation rate (**0.0412 day■¹**).
• **Growth Summary:** The observed tissue changes follow a slow biophysical reaction-diffusion pattern, consistent with post-radiation tissue inflammation rather than rapid cellular expansion.

MULTIMODAL RECURRENCE & PSEUDOPROGRESSION ANALYSIS:
Integrated machine learning analysis evaluating clinical biomarkers, timeline post-treatment, and biophysical growth dynamics yields:
• **Probability of Pseudoprogression (Treatment Effect): 78.4%**
• Probability of True Tumor Progression: 14.2%
• Probability of Stable Disease: 7.4%
• **Key Contributing Factors:** IDH mutation status, favorable MGMT methylation, and timing within 12 weeks post-radiotherapy strongly favor treatment-induced tissue reaction.

RANO 2.0 CLINICAL RESPONSE EVALUATION:
Comparison with baseline MRI scans shows an enhancing volume shift from **12.00 cm³** to **18.50 cm³** (a **+54.2%** volumetric change).
Under standard RANO 2.0 guidelines within the first 12 weeks post-radiotherapy, an enhancing expansion supported by high treatment-effect probability is classified as **Pseudoprogression (PsP)**.

IMPRESSION:

The enhancing volume shift (+54.2%) observed at 45 days post-radiotherapy in an IDH-mutated, MGMT-methylated glioblastoma patient is clinically consistent with **PSEUDOPROGRESSION (Radiation Necrosis / Treatment Effect)** rather than true disease recurrence.

CLINICAL RECOMMENDATION: Re-evaluate MRI scan in 4 to 8 weeks.